09. Business Interpretation, Results & Limitations

Translate model outputs into responsible credit decisions.

Project: CreditWise – Loan Approval Prediction

Source: supplied Loan-Approval-main.zip | Notebook: credit_wise.ipynb | Dataset: loan_approval_data.csv

Business interpretation

The notebook's simple correlation analysis shows Credit_Score as the strongest linear association with approval (about +0.451), followed by Applicant_Income (+0.120). The baseline models achieve 0.760–0.865 accuracy; after the squared-feature experiment, Logistic Regression reaches 0.880 accuracy and 0.810 F1, while GaussianNB reaches 0.811 precision. These are repository-reported figures, not independently validated estimates.

Risk and fairness considerations

- Loan decisions are high-impact. A model can systematically disadvantage groups even when aggregate accuracy looks good.
- Gender, age, marital status, property area and employer category may be sensitive or proxy attributes; legal permissibility depends on jurisdiction and product. Exclude or constrain features according to applicable law and governance policy.
- Assess selection rates, false-positive/false-negative rates, calibration and error distributions across legally appropriate groups.
- Do not infer causal reasons for a decision from correlation or raw feature importance.
- Provide adverse-action reason codes and meaningful human review where required.

Limitations in this repository

- All columns, including target, are imputed before train/test split; target labels are fabricated for 50 rows.
- Pre-split imputation and one-hot encoding create leakage.
- No stratification or cross-validation.
- Education uses LabelEncoder, imposing an arbitrary ordinal structure.
- Second feature pass drops original DTI and Credit Score after squaring them.
- No systematic tuning, probability calibration, threshold optimization or fairness audit.
- No dataset provenance, date split, units or policy context.
- No external validation and no documented final decision threshold.

Recommended next steps

- Repair the target-label issue and audit the missingness mechanism before modelling.
- Build leakage-safe preprocessing and stratified/temporal validation.
- Define asymmetric costs for false approvals and false rejections with risk/compliance stakeholders.
- Compare regularized logistic regression and tree ensembles; calibrate probabilities if used for decisions.

- Assess fairness and explainability; document permissible variables and governance.
- Validate on a separate time period or external population before any use.

Interview close

> “The first pass showed Logistic Regression and GaussianNB at 86.5% accuracy, while KNN was lower. After adding squared terms, Logistic Regression reached 88% accuracy and an F1 of about 0.81; GaussianNB had the highest precision at about 0.81. I would not present these as production-ready results because the notebook imputes missing target labels and preprocesses before splitting. Correcting those issues and validating fairness and cost-sensitive thresholds would be my next steps.”